

A Closed Loop Intent Based Agent for Resilient Multi-UPF 5G Networks



Constantinos Gkinos, Maria Christopoulou, Ioannis Vasalos, Michail Alexandros Kourtis, George Xilouris

Institute of Informatics and Telecommunications, National Centre for Scientific Research “Demokritos” (NCSRD)

We present a Closed Loop Intent Based Agent for autonomous resilience in 5G Core networks. By leveraging the Uplink Classifier (ULCL) and a custom Dynamic Steering API, our system detects “gray failure” on the N9 interface and triggers real-time path relocation. As gray failure we define a subtle performance degradation, such as high packet loss, occurring while a network link remains technically active and seemingly operational. Experimental results using free5GC [1] demonstrate that our agent can restore a failing video service from a Mean Opinion Score (MOS) of 2.0 to 3.5 in under 10 seconds, proving the viability of self-healing softwarized user planes.

Problem Statement

5G networks are vulnerable to gray failures.

Gray Failure: packet loss $\geq 10\%$ on the N9 interface causing MOS to drop below 3.5, while the link appears healthy. Threshold selected per ITU-T P.800.2 "acceptable" service quality [3].

System Design

- Free5GC to simulate the 5G network.
- Branching UPF connected to two Anchor UPFs via N9 interface [2].
- Agent on free5GC monitors N9 via periodic probing.
- If gray failure is detected, Steering API is called.
- Traffic is steered to the second Anchor UPF without interrupting the session.

Results

Metric	Phase 1: Baseline		Phase 2: Gray Failure		Phase 3: Recovery	
Injected Failure (%)	22%	97%	22%	97%	22%	97%
Packet Loss(%)	0.04%	0.13%	18.7%	47.43%	8.97%	6.47%
Avg. Inter-arrival	7.45 ms	4.53 ms	2.097 ms	1.13 ms	1.74 ms	1.56 ms
Estimated MOS	4.50	4.50	3.00	2.00	3.50	3.50
Quality	Excellent		Fair	Poor	Good	

- **MOS: 2.0** $\rightarrow$ **3.5** restored in < 10 seconds.
- **Video Quality** injecting 22% and 97% packet loss.

System Architecture & Experimental Methodology

Default path: ---
Steering path: ---
Attacker:

1. Baseline

HD RTP video from UE $\rightarrow$ Anchor UPF $\rightarrow$ Data Network
MOS 4.5 – Excellent.

2. Gray Failure

tc - netem injects 22-97% packet loss + 15 ms latency on N9 interface.
MOS drops to 2.0-3.0.

3. Recovery

Agent detects $> 10\%$ loss
Steering $\rightarrow$ Anchor UPF 2.
MOS restored to 3.5.

Future Work

- **ML detection:** replace threshold probing with machine learning detection.
- **Scalability:** larger UE populations and dynamic slice configurations.
- **Real hardware:** deploy and test on physical 5G equipment.

Acknowledgements

This work has been funded by the European Commission through SNS JU project MARE (Grant Agreement no. 101191436). Views and opinions expressed are, however, those of the author(s) only and do not necessarily reflect those of the EU or SNS JU.

References

- [1] free5GC Project. <https://free5gc.org>
- [2] 3GPP, TR 23.725 v16.0.0, 2019.
- [3] ITU-T, Rec. P.800.2, Jul. 2016.